

1 e-Kanban

Utilization

The kanban control as decentralized production control has been designed to apply the pull principle.

MES represents kanban control based on this principle in electronic form (e-kanban). Therefore, the flow of the relevant kanban objects is displayed within their closed loop (from supply to consumption and vice versa).

Prerequisites

These general definitions need to be taken into account for using e-kanban:

- Instead of physic kanban cards, MES uses or manages (electronic) kanban objects.
- The ideal total quantity of circulating kanban objects has to be calculated in advance. MES does not provide calculations or algorithms for this purpose.
- In MES the kanban objects are always posted from one material buffer to the next. It is not intended to repost them to the object "machine" or "machine group".
Consequently, supply and consumption are represented as material buffers in the system and to be configured accordingly.
- The basic package first only refers to the simple case. Here, the closed loop between one supply and one consumption is only taken into account, i.e. it is not taken into account how many machines actually pertain to the consumption and how often the material/kanban object physically moves from one "consumption buffer" to the consuming machine.

The kanban objects are not traced within the supply even if it comes to multi-level processes (kanban order includes several OPs) for replenishment production.

Procedure

Creating kanban objects

The kanban objects are created and managed in MES. Generally, a status is kept for the objects. Once the kanban object has been created, it is assigned the status "initial".

Generation of filled kanban object at the supply

The supply (material buffer) for replenishment production can be operated by one or several machines.

A machine produces the relevant material/article for a kanban order.

If a kanban object is filled, the status of the kanban object is set to "filled" by the dialog "fill kanban". However, it is irrelevant at which machine of the supply the kanban object was produced.

The kanban object is directly posted in the material buffer of the consumption defined within the closed loop.

Provision of the kanban object at the end point: consumption

The kanban object is provided at the consumption and, once it has been filled, it is directly posted into the material buffer of the consumption that is defined within the closed loop.

In MES the consumption (material buffer) is represented as:

- A general material buffer (supermarket for several machines/machine groups). In this case, the system cannot trace the kanban object (to which machine it was sent)
- The material buffer of the machine (preceding)

Emptying the kanban object at the end point: consumption

The kanban material provided at the consumption is "used/consumed" for production at a machine.

If the kanban material of a container is completely consumed, the kanban object is set to the status "empty" by the dialog "empty kanban" and directly posted onto the defined material buffer of the supply.

However, it is irrelevant to the kanban process at which machine following the consumption the container was actually emptied.

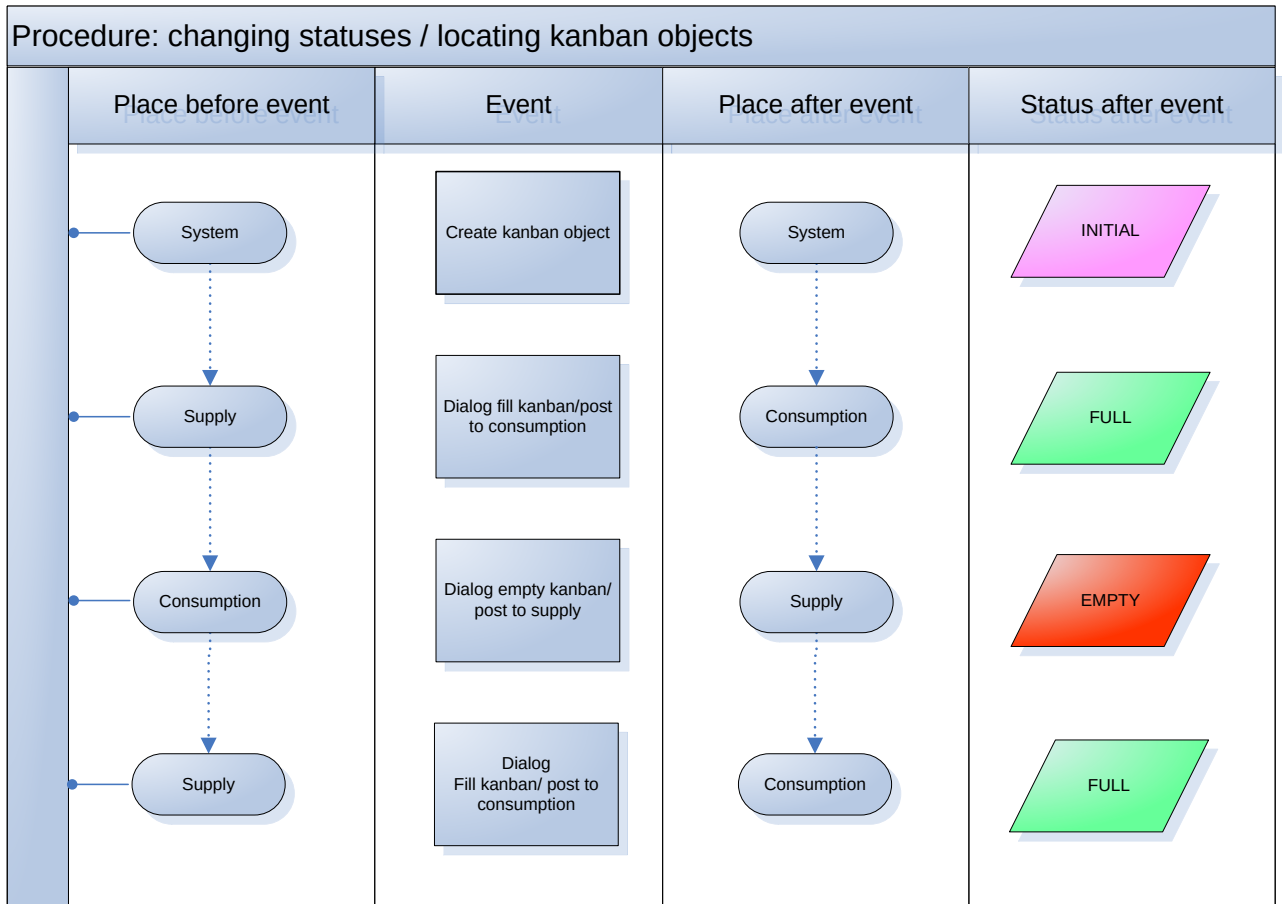
Generation of orders on request

If required, the consumption takes the kanban object from the supply. If machines are indirectly regarded as subordinate elements of the supply, replenishment production will be triggered by consuming or emptying the kanban objects (dialog: empty kanban) at the relevant machine/terminal.

The requirement of a new kanban object for this material/closed loop is visualized immediately at the supply/terminal - electronic kanban board.

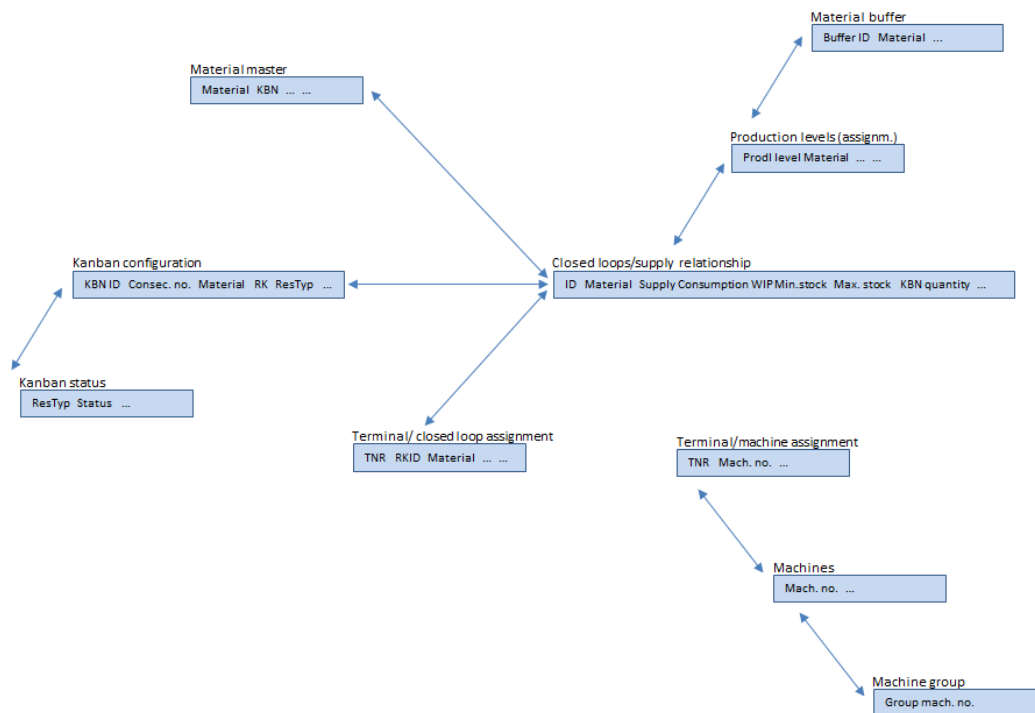
Once a minimum quantity of empty kanban objects has been reached, a planned kanban order that is shown within the order list at the supply can be generated automatically. The planned kanban order is enhanced by any further empty kanban object.

If the maximum quantity of empty kanban objects is reached, the planned kanban order becomes a fixed kanban order that can no longer be changed. Quantities are no longer updated. The fixed kanban order is also shown in the order list at the source.



Used master data and their connection

The following, cooperating master data are used to represent the kanban process within MES. They need to be created in the system.



Order generation

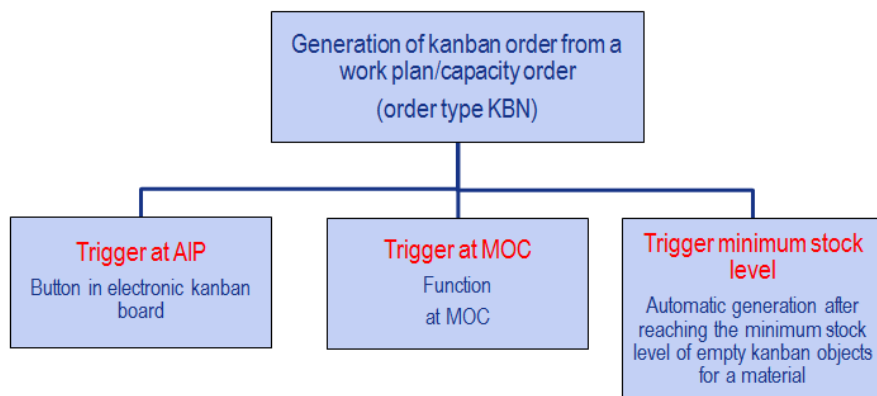
General:

In general, a kanban order is generated from a work plan created in MES or from a capacity order provided by the ERP system for the order type "KBN" (can be named flexibly). However, the generation itself might be due to several different situations.

Consequently, the following options to generate kanban orders are provided:

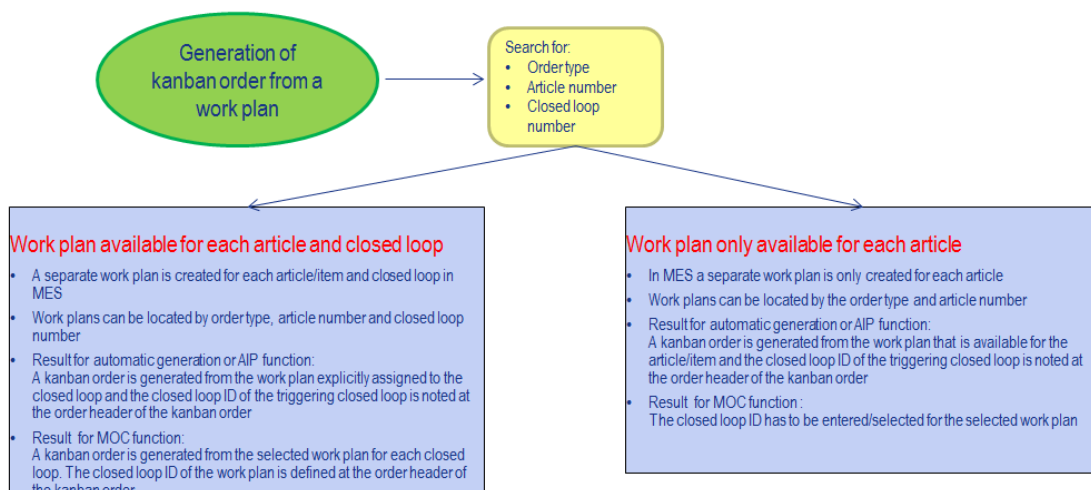
- Manually at the AIP
The dialog "electronic kanban board" provides the option to generate a kanban order for an article/material and closed loop at the push of a button.
- Manually at MOC (function of the application "kanban stock")
Orders pertaining to kanban orders are generated manually by the foreman at MOC from a separately created work plan or a provided capacity order at least for each article or article and closed loop.
- Automatic generation
The system automatically generates a planned kanban order for the relevant article from a work plan or provided capacity order when the "yellow zone" (minimum quantity of empty kanban cards) has been reached.

- Quantities are updated for the kanban order by every new, empty kanban object.
- The worker may start the generated, planned kanban order at the AIP at any time.
- To generate kanban orders, at least one separate work plan is created or capacity order is transferred from ERP for each article.
- Once the "red" zone (allowed maximum quantity of empty kanban cards) has been reached, the quantities are updated for the planned kanban order the last time and the planned kanban order becomes a fixed kanban order.

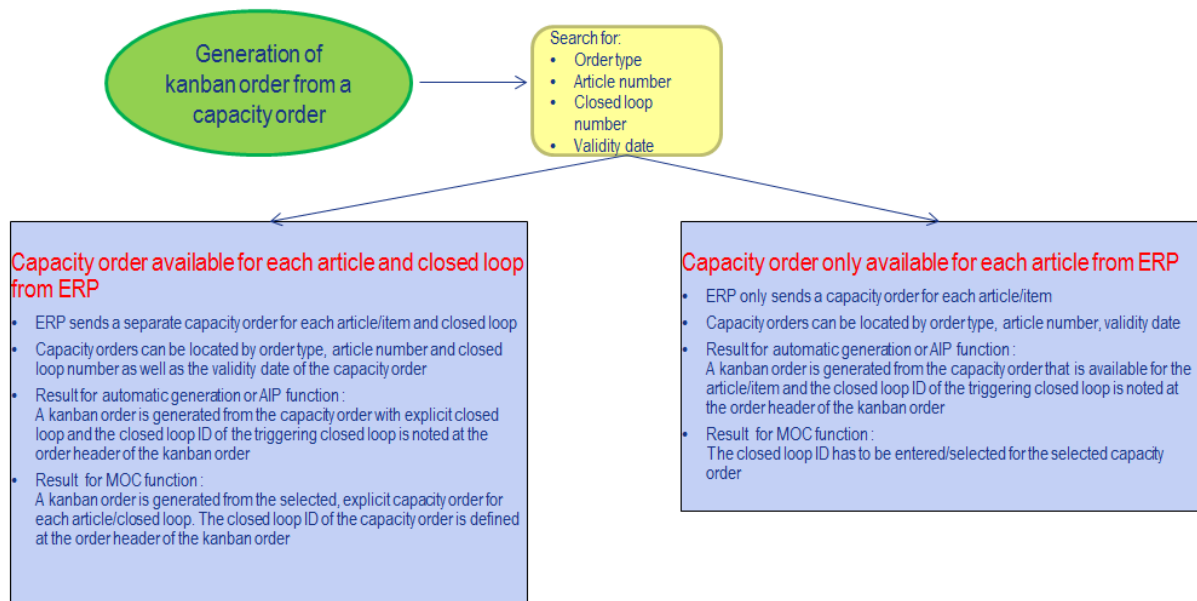


The work plans to generate kanban orders are created in MES or capacity orders are transferred from ERP. The system considers the below-mentioned situations:

Generation from a work plan:



Generation from a capacity order:



Utilization of a work plan/capacity order

- Utilization of a work plan/capacity order **for each article and supply relationship/closed loop**

Even though there might be several supply relationships/closed loops (ways from supply to consumption) for one article/item, the system has to generate a relevant kanban order for exactly one specific closed loop. The work plan/capacity order defined for this article/closed loop is used. The ID of the supply relationship/closed loop is entered in the header data to find the correct work plan / capacity order (shown in work plan header/ order header --> assignment --> master data --> supply relationship/ closed loop).

If no closed loop ID is available at the work plan/capacity order header, the first work plan/capacity order found for the article number will be used.

The closed loop number of the **triggering** closed loop is taken over for the generated kanban order.

- Utilization of a work plan/capacity order **for each article only**
 - This is sufficient, for example, if there is only one supply and consumption for each article, i.e. there is only one closed loop and no differentiation has to be made. In this case, it is not necessary to define a closed loop number for the work plan header/capacity order header.

If no closed loop ID is available at the work plan/capacity order header, the first work plan/capacity order found for the article number will be used.

For which closed loop the generated kanban order is used, results from the assignment of the closed loop ID for which the kanban order is triggered/generated (e.g. by using the relevant AIP function or the automatic generation function). The closed loop number of the triggering closed loop is taken over to the generated kanban order.

- It might be the case that there are several closed loops for each article but the same and only created work plan/capacity order is always used. Then it is not necessary to define a closed loop number for the work plan/capacity order. For which closed loop the generated kanban order is used, once again results from the assignment of the closed loop ID for which the kanban order is triggered/generated (e.g. by the relevant AIP function or the automatic generation function).
- Another possibility is to define a work plan/capacity order for each article/item only but to enter or select a specific closed loop number when generating a kanban order by using MOC functions. For which closed loop the generated kanban order is used, results from the assignment of the entered closed loop ID for which the kanban order is triggered/generated.

Prerequisites, operating conditions



- It has to be defined in advance whether a system uses capacity orders or work plans to generate kanban orders. Both variants cannot be used at the same time. The system uses the INI configuration specifying if work plans or capacity orders are to be used.
- Within the system work plans relating to articles or to articles and closed loops generally have to be defined for the order type "kanban orders" (e.g. "KBN").
- In general, the planned kanban quantities have to be defined for all kanban articles within the system.
- Number ranges for the automatic generation of the order number for kanban orders have to be created or used within the system. Number ranges can be configured flexibly and depend on the order type.

Data:

Input data to find the appropriate work plan/capacity order to "generate orders":

- Article number of the triggering article (from AIP function, MOC function, automatism)
- Closed loop ID of the triggering closed loop (from the relevant AIP function, MOC function, automatism)
- All data defined in the capacity order or work plan

The generated kanban order uses the following data from the provided work plan or capacity order:

- Scheduled start → point in time of the generation
- Scheduled end → point in time of the generation + quantity * target cycle → standard formula
- The work plan/capacity order may include several operations; however, in general it is only one OP. They are taken over for order generation and used for planning based on the default configuration.
- The quantities for the generation depend on the number of empty KBN objects and planned KBN quantities (from kanban master data) for the triggering article/closed loop and are calculated for the kanban order.
- The machine group of the OP/OPs is entered in the work plan/capacity order and planning is based on it.
- Target data (target cycle, partitioning, bill of material, etc.) is taken over from the work plan/capacity order.
- The article numbers of the order header and the last OP are identical. The article number from the order header is used for order generation. The other article numbers of the order are filled based on the work plan/capacity order.
- The closed loop ID of the triggering closed loop is used for the kanban order. The closed loop ID that is defined for the work plan/capacity order is not used.
If the kanban order is generated by the MOC function "generate order", the entered closed loop ID is used and transferred to the kanban order.

Procedure/processing

A kanban order can be generated by the following three possibilities:

- Manually at MOC
Utilization of the function "Generation of kanban orders using MOC" and generation of a kanban order from a defined work plan or capacity order.
- Manually at AIP
The "electronic kanban board" of AIP enables the generation of kanban orders for a selected article and closed loop. The order is generated from a work plan/capacity order created for this article/martial and closed loop.
- Automatic generation
If the minimum stock level of empty kanban cards (yellow zone, e.g. ≥ 5) that is configured within the supply relationships/closed loop for an article (e.g. 5 empty kanban objects) is exceeded, a planned kanban order is generated from the work plan/capacity order defined for the article and closed loop.

Once another kanban object with the status "empty" is available, the number of empty kanban containers increases and the planned kanban order is increased by the empty kanban object (e.g. the included planned kanban quantity):

- quantities are updated for the planned kanban order by every additional empty kanban object
- the worker may start the generated, planned kanban order at the AIP at any time

the order/operation status of the kanban order remains "planned" and the OP can be logged on.

Once the allowed maximum quantity of empty kanban cards ("red" zone) has been reached, the quantities are updated for the planned kanban order the last time and the planned kanban order becomes a fixed kanban order

The order/operation status of the kanban order is prepared and the OP can be logged on.

The kanban order is no longer changed by an automatic or manual quantity update.



Please note: This function can be enabled/disabled. This function cannot be used in a reasonable manner if the generation from the electronic kanban board is to be used.

The actual logic/function for generating orders is triggered by the mentioned three possibilities.

Result:

The article number and closed loop ID of the triggering closed loop or the triggering function are used as input data for all variants to find the appropriate work plan or capacity order and to generate the kanban order.

- The order or operation quantity refers to the number of currently empty kanban objects from this closed loop times the defined kanban quantity from the supply relationships/closed loops.

Order quantity/operation quantity = number of empty kanban objects x defined kanban quantity of the material

- The order type = KBN
- The order/operation status of the kanban order is prepared and the OP can be logged on
- The closed loop ID of the triggering closed loop is transferred automatically to the order header (in assignment --> master data --> closed loop) of the generated kanban order. The closed loop ID that might be defined for the work plan/capacity order is not taken over.



- A cyclic program checks for which article and closed loop the minimum stock/maximum stock of empty kanban objects has been reached and successively generates one order after the other of the kanban order.

- While a kanban order is being generated for an article/closed loop, this entry is blocked for other functions (this prevents the AIP user from generating a kanban order for the same closed loop during automatic generation).

Return values

- General
 - Shows the generated kanban orders/OP in the status "prepared" within the MOC order overview
 - Shows the generated kanban orders/OP in the status "prepared" within the AIP sequencing list (selected as fixed kanban orders/selected as planned kanban orders)
- The generated kanban order includes an assignment for the closed loop at the order header for which it was created (see *order info* → *tab order header assignment* → *master data* → *closed loop number*).

Capacity orders/ERP interface

General:

There is a general interface connection to diverse ERP systems for e-kanban (SAP-PPDC and EIS-ERP). Consequently, orders can be transferred from ERP to HYDRA and, vice versa, uploads can be posted on them (recorded durations and quantities).

The transferred orders differ from common production orders, as they are "capacity orders". In contrast to common production orders, they have the following tasks:

- Capacity orders are provided to generate planned/fixed kanban orders within MES. Within MES the capacity order is regarded as a work plan or template to create kanban orders and, therefore, it includes all relevant data required to produce operations within the order structure.
- Consequently, capacity orders are generated within ERP and transferred to MES including the BOM, PRT list and additional data (target specifications) that are determined from ERP master data.
- Normally, capacity orders are built on a monthly basis as capacitive forecast and transferred to MES.
- The planned target quantity of the capacity order is reduced by the number of actual quantities to be produced for a generated kanban order that is planned/fixed or with each quantity update for the kanban order. This is shown in MES.

- The recorded quantities and durations of a kanban operation are also posted in relation to orders on the capacity order/operation and uploaded to ERP. Thus, goods movements or consumption can be performed in a retrograde manner in ERP.

A goods movement is not generated and transferred to ERP every time a kanban object is produced.



If work plans are already used as template to generate kanban orders, **capacity orders cannot be used at the same time.**

In MES the capacity order is **not** used for capacity planning/machine assignment in HLS.

Prerequisites, operating conditions:

- Utilization of the standard ERP interface for production orders (including component list)
 - EIS-ERP
 - EIS-EZI
- Utilization of the standard SAP interface for discrete production orders (including component list, order type)
 - SAP-PPDC
 - SAP-ISS

Data within the capacity order:

These definitions have to be made or this data has to be transferred within the order interface.

ERP → MES (Download)

- Order header data
 - Order type for capacity orders = e.g. ZKAP (Please note: the order type can be configured)
 - Article for which the kanban order is to be generated
 - Closed loop for which the kanban order is to be generated (new field in ERP-EIS and SAP-ISS/function module AK)
 - Target quantity (monthly target quantity)
- Operation data
 - Article
 - Machine group
 - Flag "planned for machine group"
 - Target quantity (monthly target quantity)

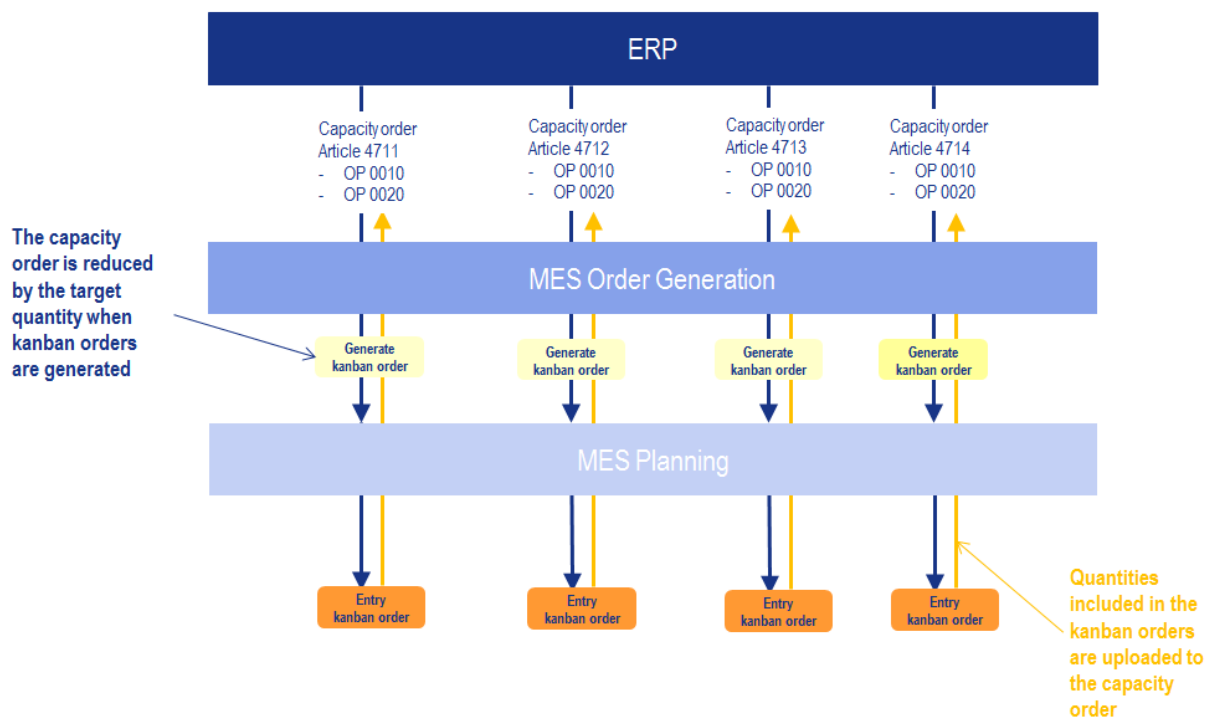
MES → ERP (uploads)

Relevant data included in uploads for capacity orders:

- Capacity order number
- Durations from kanban orders
- Quantities from kanban orders
- Please note: The machine number might not be known in ERP

Procedure/processing

Capacity orders are generated in ERP. Once they have been released, they are transferred to MES. In the first place, capacity orders are used as "templates" to generate kanban orders in MES. In general, the procedure is as follows:



Transfer of capacity orders from ERP

ERP transfers the capacity order e.g. with the order type "ZKAP" (can be configured). MES takes over the capacity order and displays it in all relevant standard applications (just as it is the case for production orders). These are, for example:

- Inbound transactions
- Backlog of orders
- Order information

In MES the capacity order is used like a work plan for an article and closed loop.

Once collected in MES, the capacity order may have the following statuses for the different objects:

- Order header → prepared
- Operations → prepared

The system may only include one valid capacity order at a time from which kanban orders can be generated. Even though several capacity orders might exist for an article/closed loop in the system.

The valid capacity order is always the newest/most recent capacity order within the system.

Utilization of capacity orders in MES

Capacity orders are used as "replacement work plan" and kanban orders are generated from them. For this reason, kanban orders are generated by using the same functions as for work plans. Triggers are:

- Manually at AIP
- Manually at MOC
- Automatic generation of orders

Generation of orders from capacity orders

Orders are always generated based on the same logic. The found capacity order (for article, closed loop, validity date) is used as template and a kanban order is generated.

The procedure for generating kanban orders from capacity orders is the same as for generating them from MES work plans.

The essential difference for the generation of kanban orders from capacity orders is that the target quantity within the capacity order is reduced by the order quantity of the kanban order every time a kanban order is generated.

With the automatic generation of planned/fixed kanban orders, the target quantity within the kanban order is increased and the target quantity within the capacity order is reduced every time quantities are updated (with every new, empty kanban object for this article and closed loop).

Uploads to capacity orders

The produced quantities and durations from the kanban order are transferred to the capacity order (added up).

For capacity orders, uploads relating to orders are also transferred to the ERP system. They are generated if a kanban order is interrupted/finished.

Status of a capacity order

The capacity order may have the following order/operation statuses (they are identical):

- Prepared: The capacity order has been transferred from ERP and created in MES.
- Started: The first kanban order has been generated for the capacity order.
- Finished: the last kanban order has been generated for the capacity order and the capacity order is no longer applicable (a newer capacity order is available).